

Original Article

Wnt/ β -catenin signaling pathway promotes abnormal activation of fibroblast-like synoviocytes and angiogenesis in rheumatoid arthritis and the intervention of Er Miao San

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ABSTRACT

Background: Er Miao San (EMS) is an important herbal formula and a representative prescription for the treatment of the downwards flow of damp-heat syndrome. Clinical practice has proven that EMS can effectively treat rheumatoid arthritis (RA). Previous studies have demonstrated that EMS regulates the functions of T cells and dendritic cells and affects the polarization of macrophages. However, it is not clear whether the inhibitory effect of EMS on RA is related to the regulation of abnormal synovial activation and angiogenesis.

Purpose: The aim of this study was to elucidate the effect and potential mechanisms of EMS on the abnormal activation and angiogenesis of fibroblast-like synoviocytes (FLSs) in RA.

Methods: The effect of EMS on rats with adjuvant arthritis (AA) and MH7A cells was examined by X-ray, haematoxylin-eosin (HE) staining, immunohistochemistry (IHC), ELISA and western blotting. Angiogenesis in AA rats was measured by a small animal ultrasound imaging system, immunofluorescence (IF) analysis and ELISA. An exchange between MH7A cells and HUVECs was induced using conditioned media that mimicked the microenvironment *in vivo*. CCK-8, western blotting, and scratch healing and Transwell migration assays were used to evaluate the effect of EMS on the Wnt/ β -catenin signaling pathway and angiogenesis in the inflammatory microenvironment of RA.

Results: Our results showed that EMS had a protective effect on AA rats. On the one hand, there was a decrease in paw swelling, the arthritis index, organ indices and proinflammatory factor levels, as well as relief of joint damage. On the other hand, blood flow, the number of immature blood vessels and proangiogenic factors were decreased. Furthermore, EMS reduced the expression of the Wnt/ β -catenin signaling pathway in the synovial tissue of AA rats and MH7A cells. In the inflammatory microenvironment of RA, the results were consistent.

Conclusion: This study demonstrated that EMS could protect against RA by inhibiting the abnormal activation and angiogenesis of FLSs, and the mechanism may be related to inhibiting the activation of the Wnt/ β -catenin signaling pathway.

Introduction

Rheumatoid arthritis (RA) is a chronic autoimmune disease

characterized by persistent synovial inflammation, anomalous synovial vascular hyperplasia and pannus formation (Nygaard and Firestein, 2020). The synovium is in a chronically inflamed state, which leads to

Abbreviations: AA, Adjuvant arthritis; Ang, Angiotensin; bFGF, Basic fibroblast growth factor; CFA, Complete Freund's adjuvant; CM, Conditioned medium; DC, Dendritic cell; EMS, Er Miao San; FLS, Fibroblast-like synoviocyte; HE, Hematoxylin-eosin; IF, Immunofluorescence; IFN, Interferon; IL, Interleukin; IHC, Immunohistochemistry; IOD, Integral optical density; RA, Rheumatoid arthritis; SD, Sprague-Dawley; SPF, Specific pathogen Free; TCM, Traditional Chinese medicine; VEC, Vascular endothelial cell; VEGF, Vascular endothelial growth factor.

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abnormal hyperplasia of synovial vessels and eventually damages the cartilage (M.Y. Zhou et al., 2021). Recent studies have revealed that pannus formation may be involved in the pathogenesis of RA (Liu et al., 2020; Tsai et al., 2021). Moreover, the inflammatory microenvironment of RA is closely related to angiogenesis (Wang et al., 2021).

Fibroblast-like synoviocytes (FLSs), vascular endothelial cells (VECs), and macrophages are cellular members of the inflammatory microenvironment of RA. Tumour-like proliferation of FLSs is a pathological factor in the pathogenesis of RA (Cheng et al., 2020; Liu et al., 2018). VECs mediate RA microangiogenesis through abnormal proliferation and migration (Jiang et al., 2022). FLSs and VECs play major roles in the promotion of synovial inflammation, bone destruction, and pannus formation in RA. Moreover, their altered biological functions are strongly associated with an imbalance in the inflammatory microenvironment of RA. On the one hand, there is a significant increase in the number of FLSs in the lesions of RA patients under pathological conditions, and these cells constantly secrete high levels of proinflammatory factors. In addition, the production of these inflammatory factors causes FLS proliferation, forming a vicious cycle that promotes the development of synovial inflammation in RA patients (A., 2000; Jia et al., 2023). On the other hand, the tumour-like proliferation of FLSs leads to excessive hyperplasia and hypertrophy of normal synovial tissue. Then, oxygen consumption increases, and synovial tissue experiences acute ischaemia and hypoxia, causing excessive production of synovial microvessels. Ultimately, the newly formed microvessels can provide large amounts of nutrients to mediate the growth of synovial tissue. In brief, abnormal proliferation of FLSs and angiogenesis of VECs synergistically promote the formation of the pannus, which is characteristic of RA (Chen et al., 2022; Deng et al., 2021).

The Wnt signaling pathway is a complex regulatory network; Wnt/ β -catenin is considered the classic Wnt signaling pathway, and it is involved in regulating essential physiological processes such as embryonic development, cell proliferation and differentiation (Huang et al., 2019). When the Wnt/ β -catenin signaling pathway is activated, β -catenin is transferred from the cytoplasm to the nucleus (Selvarathinam et al., 2023). It then activates the downstream target vascular endothelial growth factor (VEGF), thereby exerting its pathological effects. VEGF, which is mainly secreted by FLSs, is a potent angiogenic growth factor that is abundantly expressed in the synovium of RA patients (Fromm et al., 2019). VEGFA is widely believed to be a major mediator of angiogenesis, and its secretion is strongly associated with metastasis and poor prognosis in different cancers (Yang et al., 2022). Activation of the Wnt/ β -catenin signaling pathway could initiate the transcription of VEGFA; thus, promotion of tumour neovascularization has been a research hotspot worldwide in recent years. In some cancers, VEGFA has been shown to be a target gene of β -catenin (Sui et al., 2017). Several studies have shown that Wnt/ β -catenin signaling pathway factors are aberrantly expressed in a variety of autoimmune diseases. In animal models, altering the activation of Wnt/ β -catenin is effective (Giacoppo et al., 2016; Hauck et al., 2012; MacLauchlan et al., 2017; Sorcini et al., 2017). Moreover, the expression of β -catenin was markedly higher in RA synovial samples than in patients with osteoarthritis and general trauma (Xiao et al., 2011). The Wnt/ β -catenin signaling pathway might play an important role in the pathogenesis of RA. However, the role of β -catenin and VEGFA in the development of RA has not been reported and deserves further investigation.

Danxi Xinfu is a comprehensive medical book that comprehensively reflects Zhu Danxi's viewpoint of "Yang is always in excess and Yin is always in deficiency" and his rich clinical experience, and systematically introduces his therapeutic insights and effective formulas for fire-heat syndrome. According to traditional Chinese medicine (TCM), wind, cold, damp and heat are important pathogenic factors in RA, which is a type of blockage syndrome. Er Miao San (EMS) is an important herbal formula and a representative prescription for the treatment of downwards flow of damp-heat syndrome, which is included in the Synopsis of the Danxi Xinfu. It can clear heat and dry dampness. Clinical practice has

proven that EMS can effectively treat RA. Our group previously reported that EMS had an inhibitory effect on RA, and its mechanisms mainly involved regulating the function of T cells and dendritic cells (DCs) and affecting the polarization of macrophages (Dai et al., 2020; Ding et al., 2020; Liu et al., 2022). However, whether the inhibitory effect of EMS on RA is related to the regulation of abnormal synovial activation and angiogenesis has not been clarified.

Based on previous work, we investigated the effect of EMS on the Wnt/ β -catenin signaling pathway during the abnormal activation and angiogenesis in synovial tissue to provide new ideas for exploring anti-RA drugs by targeting synovial function and Wnt/ β -catenin.

Materials and methods

Drugs and reagents

Wnt5a/b (Cat. 2530) and p-GSK-3 β (Ser9) (Cat. 9323) were purchased from Cell Signaling Technology (Boston, USA). β -catenin (Cat. M24002) and APC (Cat. T55641) were obtained from ABMART (Shanghai, China). VEGFA (Cat. 19,003-1-AP) and secondary antibodies (Cat. SA00001-1, Cat. SA00001-2) were purchased from Proteintech (Wuhan, China). A rat Ang-II ELISA kit (Cat. E-EL-R1430c) and a rat bFGF ELISA kit (Cat. E-EL-R0091c) were obtained from Elabscience (Wuhan, China). A rat IFN- γ ELISA kit (Cat. EK380) and a rat IL-17A ELISA kit (Cat. EK317) were obtained from Multisciences (Hangzhou, China). Adavivint (SM04690) (Cat. S8761) was purchased from Selleck (Shanghai, China). Methotrexate (MTX) was provided by the First Affiliated Hospital of University of Science and Technology of China (Hefei, China).

Plant materials

EMS (Anhui Puren Herbal Pieces Co., Ltd.) is composed of *Atractylodes Rhizoma* (Cangzhu, specimen number 1,902,120,322), the rhizome of *Atractylodes lancea* (Thunb.) DC. (Compositae), and *Phellodendri Cortex* (Huangbai, specimen number 1,901,200,062), and the bark of *Phellodendron chinensis* Schneid (Rutaceae). The samples were identified by Dr. Liu SJ (School of Pharmacy, Anhui University of Chinese Medicine), and one specimen of each (ID: EMS-19-01) was preserved in the Herbarium of Pharmacy, School of Pharmacy, Anhui University of Traditional Chinese Medicine (Hefei, China).

The mixture of *Atractylodes Rhizoma* and *Phellodendri Cortex* was dried and crushed, soaked for approximately 2 h, and then extracted with 10, 8 and 6 times water for 1.5 h, 1 h and 0.5 h, respectively. The filtrate was combined and concentrated at 60 °C. After five extractions with equal volumes of petroleum ether and ethyl acetate, the ethyl acetate fraction of EMS was concentrated to the experimentally required doses (calculated using the crude drug). UPLC analysis of the EMS extracts is shown in Supplementary Fig. 1.

Modelling and treatment

All animal experiments were performed at the School of Pharmacy, Anhui University of Traditional Chinese Medicine (Hefei, China). The experimental protocol was approved by the Experimental Animal Ethics Committee and adhered to the guidelines for animal care and use on 21 December 2020 (AHUCM-rats-2,020,016). Male specific pathogen-free (SPF)-grade Sprague-Dawley (SD) rats were acquired from the Animal Department of Anhui Medical University, China, and housed in standard laboratory conditions (under a controlled temperature of 22–26 °C with a 12-h light and 12-h dark period).

The SD rats were acclimatized and fed for one week and randomly divided into a blank control group and an EMS group (3 g/kg). Physiological saline and EMS (0.3 g/ml) were administered by gavage at a dose of 1 ml/100 g once/d for 7 d 1 h after the last dose, the rats were anaesthetized with pentobarbital sodium, and whole blood was

collected from the abdominal aorta. Serum was collected after centrifugation at 3000 r/min for 15 min and inactivated in a water bath at 56 °C for 30 min. Finally, the serum was filtered through a 0.22 µm microporous membrane and stored at −20 °C. MS analysis of the serum is shown in Supplementary Fig. 2 and Supplementary Table 1.

The rats were randomly divided into 5 groups: normal, model, EMS (1.5 g/kg), MTX (0.5 mg/kg), and SM04690 (0.6 µg/25 µl). The adjuvant arthritis (AA) model was induced by intradermal injection of 0.1 ml of complete Freund's adjuvant (CFA) into the right hindfoot plantar of the rats. After 17 d, the drugs were given. The EMS (1.5 g/kg) group was administered the treatment by gavage once *per* day for a total of 14 times; the MTX (0.5 mg/kg) group was administered the treatment by gavage every three days for a total of 5 times; and the SM04690 (0.6 µg/25 µl) group was intra-articularly injected once after successful modelling.

Evaluation of arthritis

The volume of the left hind paw (secondary side) of the rats was measured by a PV200 vol metre before modelling. After successful modelling, the volume of the secondary paw was measured every three days. Paw swelling = Postimmunization paw swelling – Preimmunization paw swelling.

Beginning on day 18 after immunization, two independent researchers who were unaware of the treatment plan scored each group of rats on the arthritis index every three days, and the scoring criteria are shown in Supplementary Table 2. The degree of lesions in the extremities of each rat was accumulated and scored as the polyarthritis index, with a maximum of 16 points *per* rat.

Organ indices

The rats in each group were weighed before the end of the experiment. Afterwards, the thymus and spleen were removed, weighed and recorded. Eventually, the thymus index (thymus weight/body weight) and spleen index (spleen weight/body weight) were calculated.

X-ray analysis

The ankle joints were fixed in 4% paraformaldehyde. The samples were placed in the animal X-ray irradiator under the following conditions: image resolution of 15 LP/min, photographic conditions of 22.0 KV 4 mA, and 0.1 mGy. Swelling in joint soft tissue, joint deformation and the joint space were observed.

Haematoxylin-eosin (HE) staining

The knee joints of rats were fixed with 4% paraformaldehyde for 72 h, decalcified with 10% EDTA-2Na for 30 days, dehydrated, cleared, embedded in paraffin, sliced (5 µm), and stained with HE. Two independent researchers observed and recorded histopathological changes in the knee joint using an optical microscope. Synovial proliferation, cellular infiltration, and pannus formation. were assessed according to the scoring criteria in Supplementary Tables 3–5.

Immunohistochemistry (IHC)

Paraffin sections of synovial tissue were prepared, dehydrated and incubated overnight with β-catenin antibodies at 4 °C. Goat anti-mouse IgG secondary antibodies were used, and DAB colouration and sealing were performed. Positive expression of β-catenin in the synovium was observed under a microscope.

Immunofluorescence (IF) staining

Paraffin sections of synovial tissue were prepared, dehydrated and

incubated overnight with CD31 and αSMA antibodies at 4 °C, incubated with secondary antibodies for 1 h, sealed, and observed and photographed under a fluorescence microscope within 2 h, and the integral optical density (IOD) of positive staining of mature CD31⁺/αSMA⁺ vessels, immature CD31⁺/αSMA[−] vessels and total vessels (sum of mature CD31⁺/αSMA⁺ vessels and immature CD31⁺/αSMA[−] vessels) was determined to reflect the number of blood vessels.

Preparation of conditioned medium (CM)

MH7A cells or HUVECs were cultured in culture flasks, and when the cells were approximately 80% confluent, the medium was discarded. Then, medium containing a low level of FBS was added and incubated for 24 h, and the supernatant was collected. Afterwards, the supernatant was centrifuged at 1800 rpm for 5 min, filtered through a 0.22 µm microporous membrane and stored at −20 °C. FBS was added so that the CM contained 10% FBS before use. The CM that was used in follow-up experiments was prepared according to this method.

Cell viability assay

Cells were treated according to the experimental needs, and CCK-8 (10 µl) was added to each well and incubated for 2 h. Then, the absorbance (450 nm) was determined using a microplate reader. The viability of the cells was calculated based on the OD values.

Western blotting

Total proteins were extracted with RIPA lysis buffer, or nuclear proteins were extracted according to the instructions of the nuclear protein extraction kit. Protein concentrations were determined by BCA kits. Equal amounts of protein samples were added to each well for SDS-PAGE and then transferred to PVDF membranes. After being sealed with 5% skim milk at room temperature for 2 h, the membranes were incubated with β-actin, GAPDH, β-catenin, APC and VEGFA at 4 °C overnight and exposed to secondary antibodies at room temperature for 2 h. Protein bands were visualized using ECL kits. ImageJ software was used to analyse the grey value of each band, and the control protein was used to normalize the grey values of the target protein for statistical analysis.

ELISA

The blood was collected from the abdominal aorta and centrifuged (3000 rpm for 15 min). The supernatant was collected and stored at −80 °C. The levels of interferon (IFN)-γ, interleukin (IL)-17, basic fibroblast growth factor (bFGF) and angiotensin (Ang)-2 were evaluated by ELISA according to the manufacturer's instructions. The absorption was measured at 450 nm.

Wound-healing assay

HUVECs were cultured in 6-well plates (2 × 10⁵/ml) for adherence. When the HUVECs were almost confluent, sterile toothpicks were used to gently scrape a line in the cells in each well (0 h). The HUVECs were washed to remove cellular debris and further cultured in serum-free medium with different concentrations of EMS or SM04690 for 24 h, with/without stimulation with MH7A CM for 24 h according to the groupings. Images were taken at 0 and 48 h, and the cells were observed under an inverted microscope and photographed. The wound area was measured by ImageJ software to quantify the wound healing.

Transwell migration assay

Transwell chambers were placed in 24-well plates. A HUVEC suspension with a density of 4 × 10⁴/ml was prepared. Then, 200 µl of the cell suspension was added to the upper chamber, 600 µl of MH7A CM

was added to the lower chamber, and the cells were cultured in a 37 °C, 5% CO₂ incubator for 24 h. The corresponding drug solutions in volumes of 200 µl and 600 µl were added to the upper chamber and lower chamber, respectively, and incubated for 24 h. Then, the drug solution was removed, and cotton swabs were used to gently remove the un-migrated HUVECs in the upper chamber. The cells on the lower surface of the membrane were fixed with 4% paraformaldehyde for 15 min, stained with crystal violet for 30 min, and washed with PBS until the rinse was colourless. Three randomly selected areas were photographed using an optical microscope, and ImageJ software was used to calculate the number of HUVECs within each field.

Statistical analysis

Statistical analysis was performed with GraphPad Prism V.8.0.2 (GraphPad, USA). Data in the figures are expressed as the mean ± SD, and one-way ANOVA was used to determine the differences between experimental groups. A value of $p < 0.05$ was considered statistically significant.

Results

EMS alleviated arthritis-related indices in AA rats

A previous study by our group showed that EMS (0.75 g/kg, 1.5 g/kg, 3.0 g/kg) had a significant dose–response relationship in the treatment

of AA rats (Liu et al., 2022). In the current study, a suboptimal dose of EMS (1.5 g/kg/d) was administered to AA rats for 14 consecutive days, and the overall effect of EMS was observed. To examine the effect of EMS on RA, we established an AA rat model (Day 0) to simulate RA. Another group of rats was administered SM04690, an inhibitor of the Wnt/β-catenin signaling pathway. After the successful establishment of the model (Day 17), the groups were randomized. EMS (1.5 g/kg), MTX (0.5 mg/kg), and SM04690 (0.6 µg/25 µl) were administered to AA rats from day 18 to day 33 (Fig. 1A).

As shown in Fig. 1B, AA rats had obvious redness and swelling in the paws, and nodules appeared on the tail on day 33. Moreover, paw swelling and the arthritis index in model rats increased significantly and peaked at approximately day 24. However, redness, nodules, paw swelling and the arthritis index were alleviated in the EMS, MTX and SM04690 groups (Fig. 1C, D). The thymus index and spleen index are important indicators of immune function (Li et al., 2020). The results showed that the thymus index and spleen index were increased significantly in the model group, while the increases in the thymus index and spleen index were prevented by EMS, MTX or SM04690 treatment (Fig. 1E). These results suggest that EMS has a protective effect on immune function in AA rats.

Ankle joint lesions were examined by X-ray. In the model group, obvious swelling, deformation and breakage of knee joint cartilage were observed. After treatment, swelling and deformation of the ankle joint were improved, and the joint was more intact (Fig. 1F). Pathological changes in the knee joint were observed by HE staining.

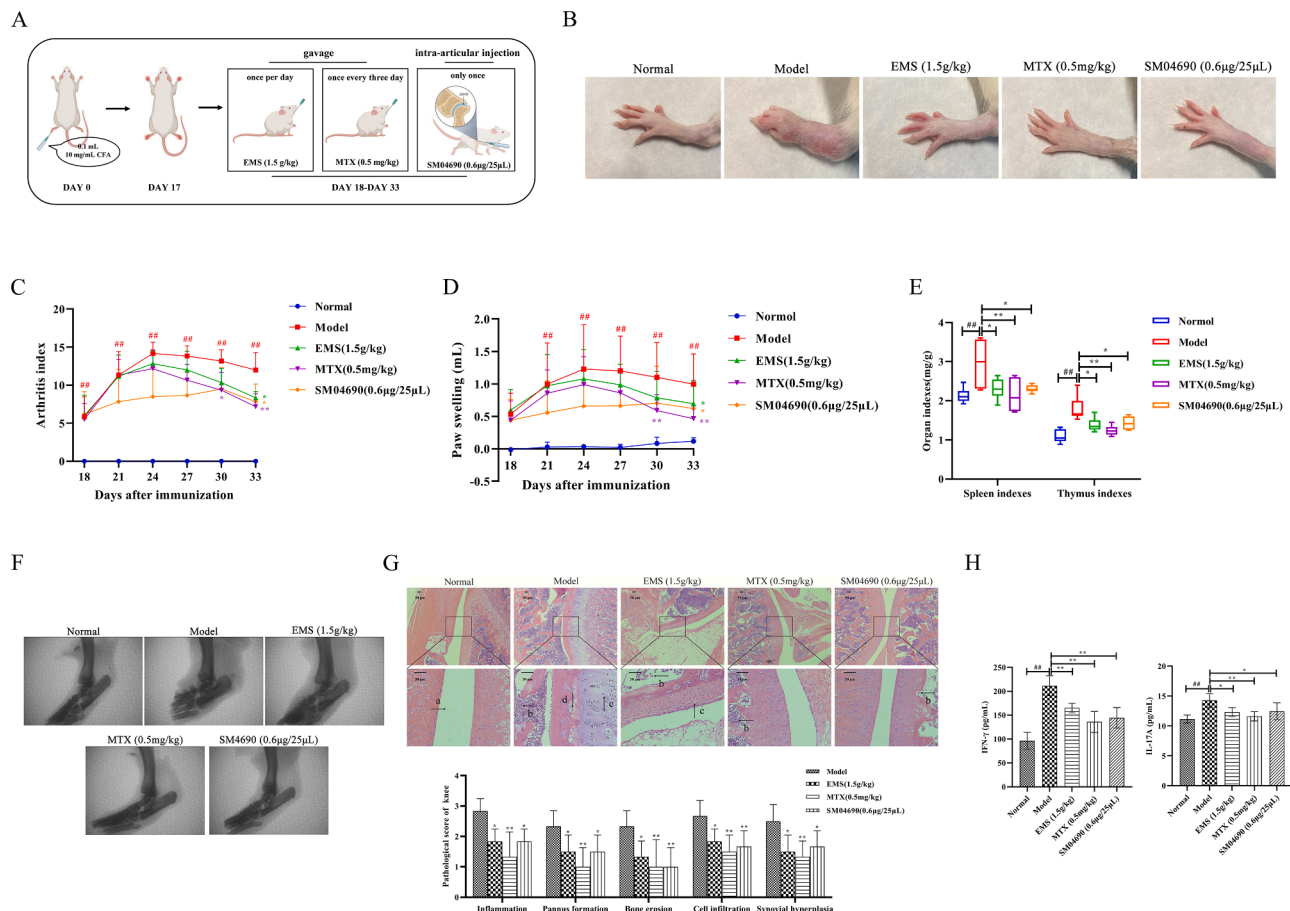


Fig. 1. EMS alleviated arthritis-related indices in AA rats. (A) Timeline of AA development and treatment. (B) Representative images of rat paw. Effects of EMS on (C) the arthritis index, (D) paw swelling and (E) organ indices in rats with AA. (F) X-ray images of the ankle joint. (G) Histological section shows synovial hyperplasia, (H) pannus formation, (I) cell infiltration, (J) synovial hyperplasia, and (K) pannus formation. (H) ELISA kits were used to evaluate the levels of IFN-γ and IL-17A in the serum samples of AA rats. The data are presented as the mean ± SD ($n = 6$). $^{##}p < 0.01$ vs. the normal group; $^{*}p < 0.05$, $^{**}p < 0.01$ vs. the model group.

Histopathological analysis showed that the knee joints in the model group exhibited significant synovial hyperplasia, pannus formation, cellular infiltration and even bone erosion. In contrast, these pathological changes were alleviated to varying degrees by EMS, MTX and SM04690 treatment (Fig. 1G). These results suggest that EMS can improve pathological changes in the joints of AA rats.

The levels of the inflammatory factors IFN- γ and IL-17A in rat serum under inflammatory conditions determined by ELISA. The results showed that the levels of IFN- γ and IL-17A were significantly increased, while EMS, MTX and SM04690 decreased the levels of IFN- γ and IL-17A (Fig. 1H). These results suggested that EMS could reduce inflammation in AA rats.

These results also showed that SM04690, which is an inhibitor of the Wnt/ β -catenin signaling pathway, could replicate the effects of EMS and attenuate arthritic symptoms and immune inflammatory responses in AA rats, suggesting that the Wnt/ β -catenin signaling pathway is involved in the pathogenesis of RA.

EMS inhibited abnormal synovial activation via the Wnt/ β -catenin signaling pathway

We further explored the mechanism by which the Wnt/ β -catenin signaling pathway is involved in abnormal RA function. The IHC results showed that there was a small amount of β -catenin staining (brown) in synovial tissue in the normal group, and the expression of β -catenin in AA rats was higher than that in normal rats. Moreover, the staining was

mainly localized in the cytoplasm and nucleus. After EMS, MTX or SM04690 treatment, β -catenin staining was differentially negative or weakly positive (Fig. 2A). Moreover, the expression of β -catenin in AA rats was verified at the protein level by western blotting, and the results were consistent with the IHC results (Fig. 2B). The results showed that EMS reduced the levels of β -catenin and VEGFA in the synovial tissue of AA rats.

To verify this finding, we carried out *in vitro* experiments. Based on the CCK-8 results, 625 nM SM04690 treatment for 24 h was used for subsequent experiments (Fig. 2C). TNF- α is a key regulator of joint inflammation and injury and can activate a variety of pathways to induce inflammation (Armaka et al., 2022). In this study, an MH7A inflammation model induced by TNF- α was established *in vitro*. The results in Fig. 2D and E show that compared with those in the control group, the expression of the Wnt/ β -catenin signaling pathway proteins β -catenin, VEGFA, p-GSK3 β (Ser 9), Wnt5a/b, and APC and nuclear β -catenin levels were markedly increased in the TNF- α group. Compared with the TNF- α group, the EMS and SM04690 groups had decreased expression of these proteins and nuclear β -catenin protein to different degrees. These findings suggest that EMS reduces APC levels by down-regulating Wnt levels, which in turn inhibits GSK3 β phosphorylation, promotes GSK3 β to add phosphate groups to serine/threonine residues at the N-terminal end of β -catenin, accelerates β -catenin degradation via ubiquitin, and reduces β -catenin expression in the nucleus, ultimately reducing VEGFA protein levels.

To further verify these results, β -catenin was upregulated with LiCl,

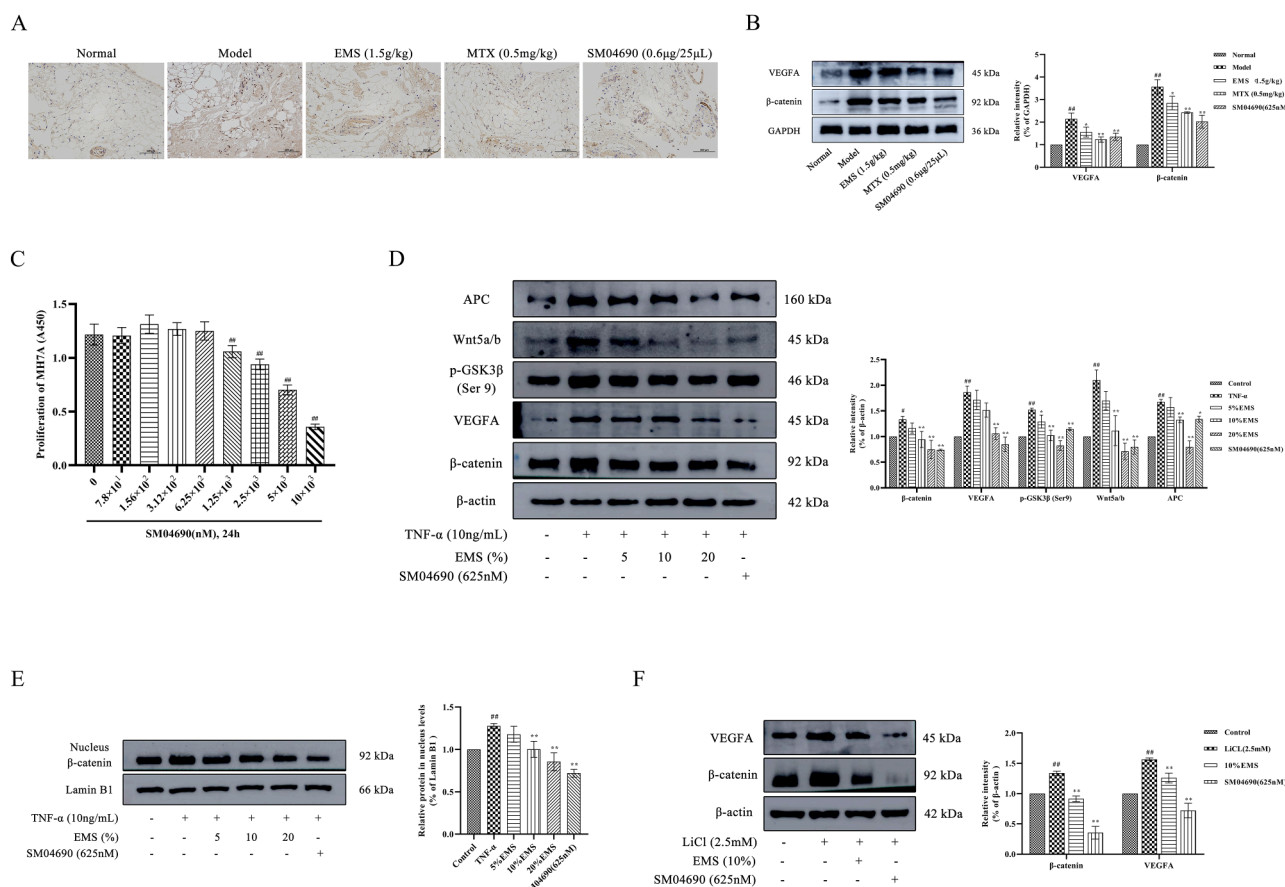


Fig. 2. EMS inhibited abnormal synovial activation via the Wnt/ β -catenin signaling pathway. IHC (A) and western blotting (B) were used to detect the expression of β -catenin and VEGFA in the synovial tissues of AA rats ($n = 3$). (C) The proliferation of MH7A cells treated with different concentrations of SM04690 for 24 h was tested using the CCK-8 assay ($n = 6$). The protein levels of (D) β -catenin, VEGFA, p-GSK3 β (Ser 9), Wnt5a/b, and APC and (E) β -catenin levels in the nucleus in TNF- α -stimulated MH7A cells were determined by western blotting ($n = 3$). (F) The protein levels of β -catenin and VEGFA in LiCl-stimulated MH7A cells were determined by western blotting ($n = 3$). The data are presented as the mean $\pm$ SD. $^{\#}p < 0.05$, $^{##}p < 0.01$ vs. Normal or Control group; $^{*}p < 0.05$, $^{**}p < 0.01$ vs. Model, TNF- α , or LiCl group.

an activator of the Wnt/ β -catenin signaling pathway, to observe its effect on VEGFA. Different concentrations of LiCl (0, 2.5, 5, 10, 20, and 40 mM) were incubated with MH7A cells for 24 h. The effect of LiCl on MH7A cells was observed by inverted microscopy, CCK-8 assays and western blotting (Supplementary Fig. 3). Thus, 2.5 mM LiCl and 24 h were selected as the concentration and time for the follow-up experiment. The western blot results in Fig. 2F show that the protein expression of β -catenin and VEGFA was markedly increased in the LiCl group compared with the control group. Compared with the LiCl group, the 10% EMS and SM04690 (625 nM) groups exhibited markedly reduced protein expression levels of β -catenin and VEGFA. These results suggested that LiCl could increase the protein expression of β -catenin, which could upregulate VEGFA expression. EMS downregulated the level of VEGFA by inhibiting the expression of β -catenin.

In short, EMS has the same effect as SM04690 and inhibits the related indices of the Wnt/ β -catenin signaling pathway in the synovial tissue of AA rats and MH7A cells. EMS might also decrease the level of β -catenin and reduce the entry of β -catenin into the nucleus, thereby decreasing the expression of VEGFA. However, VEGFA is an important indicator of angiogenesis, and we hypothesized that EMS mediated angiogenesis in RA.

EMS inhibited angiogenesis in AA rats

Ultrasound is an indispensable imaging tool in the clinical diagnosis and treatment of RA and can be used to identify a variety of common joint lesions in RA patients (Filippucci et al., 2019). The small animal

ultrasound imaging system showed that blood flow in the knee joint in AA rats was increased, while that in the EMS, MTX and SM04690 groups was decreased (Fig. 3A). These findings suggested that a pannus formed in the synovium in AA rats, and the drugs had a significant effect on angiogenesis in AA rats. To detect angiogenesis in the synovial tissue of inflamed joints, the expression of CD31 and α SMA in synovial tissue was analysed by IF staining. As shown in Fig. 3B, positive staining for numerous immature vessels was seen in the synovial tissue of AA rats, while EMS affected the formation of immature microvessels, which showed CD31⁺ staining rather than α SMA⁺ staining. Quantitative analysis showed that EMS, MTX and SM04690 could significantly reduce the number of immature CD31⁺/ α SMA⁺ vessels and total vessels in the synovium of AA rats but had little effect on the number of mature CD31⁺/ α SMA⁺ vessels. These results suggest that EMS has an anti-angiogenic effect on RA. In the inflammatory state, the secretion of proangiogenic factors in the synovial tissue of RA patients is increased, which induces neovascularization (Lim et al., 2022). Serum was collected from rats in each group, and the levels of the angiogenic factors bFGF and Ang-2 were detected. Compared with those in the normal group, the levels of bFGF and Ang-2 in the model group were significantly increased, and the levels of bFGF and Ang-2 were significantly decreased after drug treatment (Fig. 3C). These results suggested that EMS could reduce the level of angiogenesis in AA rats.

In summary, angiogenesis is involved in the pathogenesis of AA in rats, and EMS can inhibit angiogenesis in AA rats.

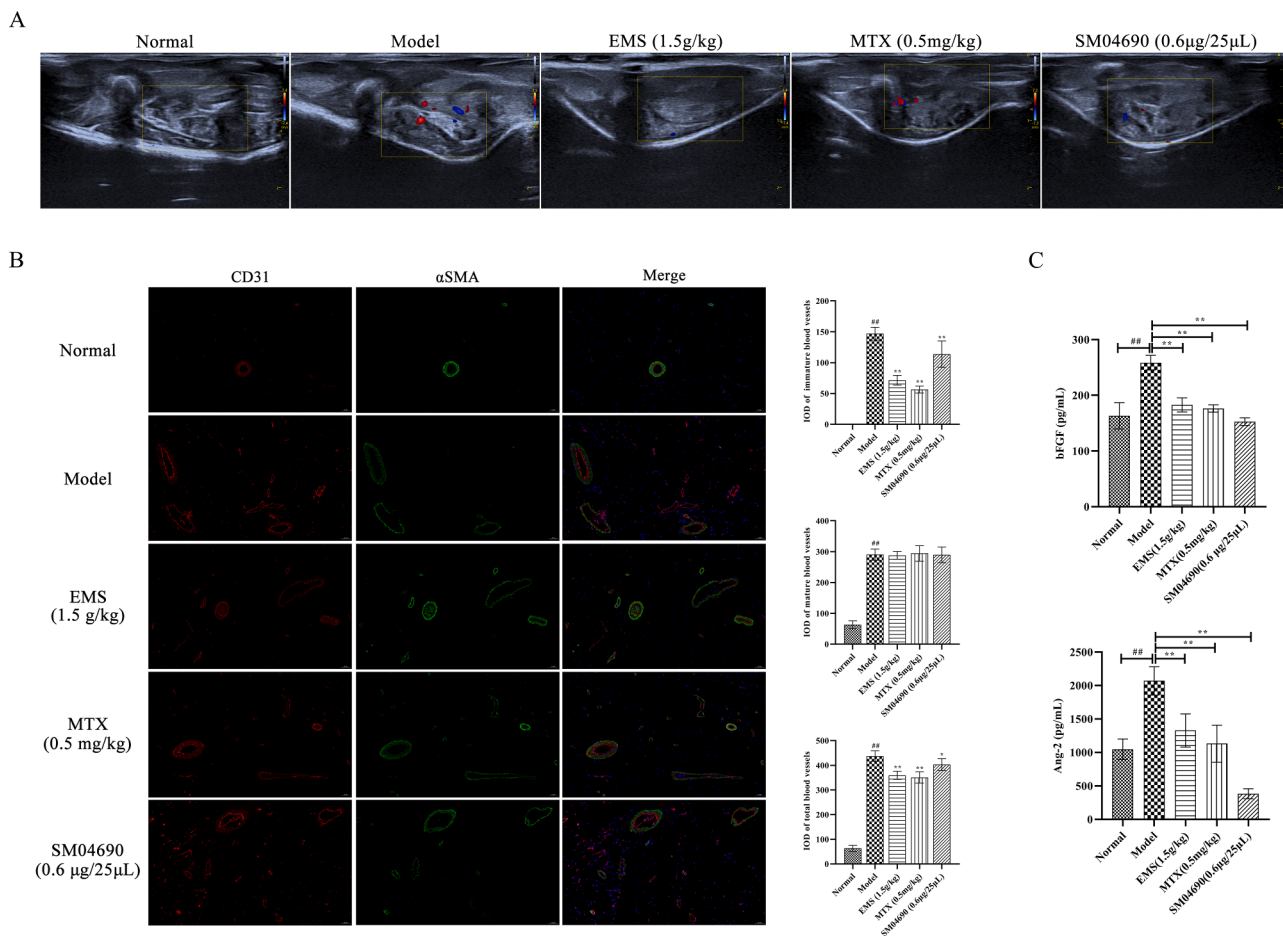


Fig. 3. EMS inhibited angiogenesis in AA rats. (A) Blood flow in the knee joint of rats was observed by ultrasound. (B) IF analysis was used to test the effects of EMS on immature vessels, mature vessels and total vessels. (C) The expression of bFGF and Ang-2 in the serum samples of AA rats was detected by ELISA. The data are presented as the mean $\pm$ SD ($n = 6$). ^{##} $p < 0.01$ vs. the normal group; ^{*} $p < 0.05$, ^{**} $p < 0.01$ vs. the model group.

EMS inhibited changes in the biological functions of MH7A cells and HUVECs in an inflammatory microenvironment

Indirect coculture is an effective technique to examine the effect of one cell on the biological function of another cell *in vitro*. The effects of different concentrations of HUVEC CM (0%, 12.5%, 25%, 50%, and 100%) on the viability of MH7A cells were observed to determine the best coculture conditions. The CCK-8 assay showed that 12.5% HUVEC CM promoted the proliferation of MH7A cells most significantly (Fig. 4A). Based on this finding, 12.5% HUVEC CM was determined to be the best concentration for the MH7A-HUVEC model in subsequent experiments. The CCK-8 results showed that 5% EMS, 10% EMS, 20% EMS and 625 nM SM04690 could reduce the proliferation of MH7A cells (Fig. 4B). Compared with those in the control group, the protein expression of β -catenin and VEGFA and the levels of β -catenin protein in the nucleus were significantly increased by HUVEC CM stimulation. The EMS and SM04690 groups had significantly decreased β -catenin, VEGFA and nuclear β -catenin protein expression levels (Fig. 4C, D).

HUVEC activation plays a key role in angiogenesis in RA. The CCK-8 assay showed that 12.5% and 25% MH7A CM promoted the proliferation of HUVECs, but 100% MH7A CM inhibited the viability of HUVECs (Fig. 5A). Furthermore, we found that 5% EMS, 10% EMS, 20% EMS and 625 nM SM04690 reduced the proliferation of HUVECs (Fig. 5B). The EMS and SM04690 groups had significantly decreased β -catenin and VEGFA protein expression levels (Fig. 5C). These findings suggested that the crosstalk between MH7A cells and HUVECs could activate the Wnt/ β -catenin signaling pathway and participate in microangiogenesis, and EMS could inhibit angiogenesis in the inflammatory microenvironment by inhibiting the expression of β -catenin, thereby reducing the entry of β -catenin into the nucleus and downregulating the expression of VEGFA.

The occurrence and development of angiogenesis is based on the biological behaviour of endothelial cells, and the migration of endothelial cells is the basis of their biological behaviour (Mahmoud et al., 2022). We further analysed the effect of EMS on the horizontal and vertical migration of HUVECs stimulated by MH7A CM. The results of the scratch healing and Transwell migration assays showed that HUVEC migration was significantly increased. The horizontal and vertical migration rates of HUVECs in the 5% EMS, 10% EMS, 20% EMS and SM04690 groups were significantly lower than those in the CM group (Fig. 5D, E). These results suggested that EMS could inhibit angiogenesis in not only AA rats but also HUVECs in an inflammatory microenvironment.

In summary, the crosstalk between MH7A cells and HUVECs could activate the Wnt/ β -catenin signaling pathway, and EMS could inhibit angiogenesis in the inflammatory microenvironment by inhibiting the expression of β -catenin, thereby downregulating the expression of VEGFA and weakening the migration of HUVECs. These results suggested that EMS could inhibit angiogenesis in not only AA rats but also the inflammatory RA microenvironment *in vitro*.

Discussion

RA is a typical autoimmune disease characterized by the formation of synovial microvessels. AA is a classic animal model of RA that is used to study the pathogenesis of RA and the therapeutic effects of drug intervention. Previous studies have shown that EMS suppresses the inflammatory response and iNOS expression by inhibiting MAPK activation and the NF- κ B signaling pathway (Chen et al., 2014). A combination of network pharmacology and experimental validation showed that EMS exerted anti-inflammatory effects by inhibiting the release of inflammatory cytokines (Guo et al., 2022). In this study, we showed that EMS

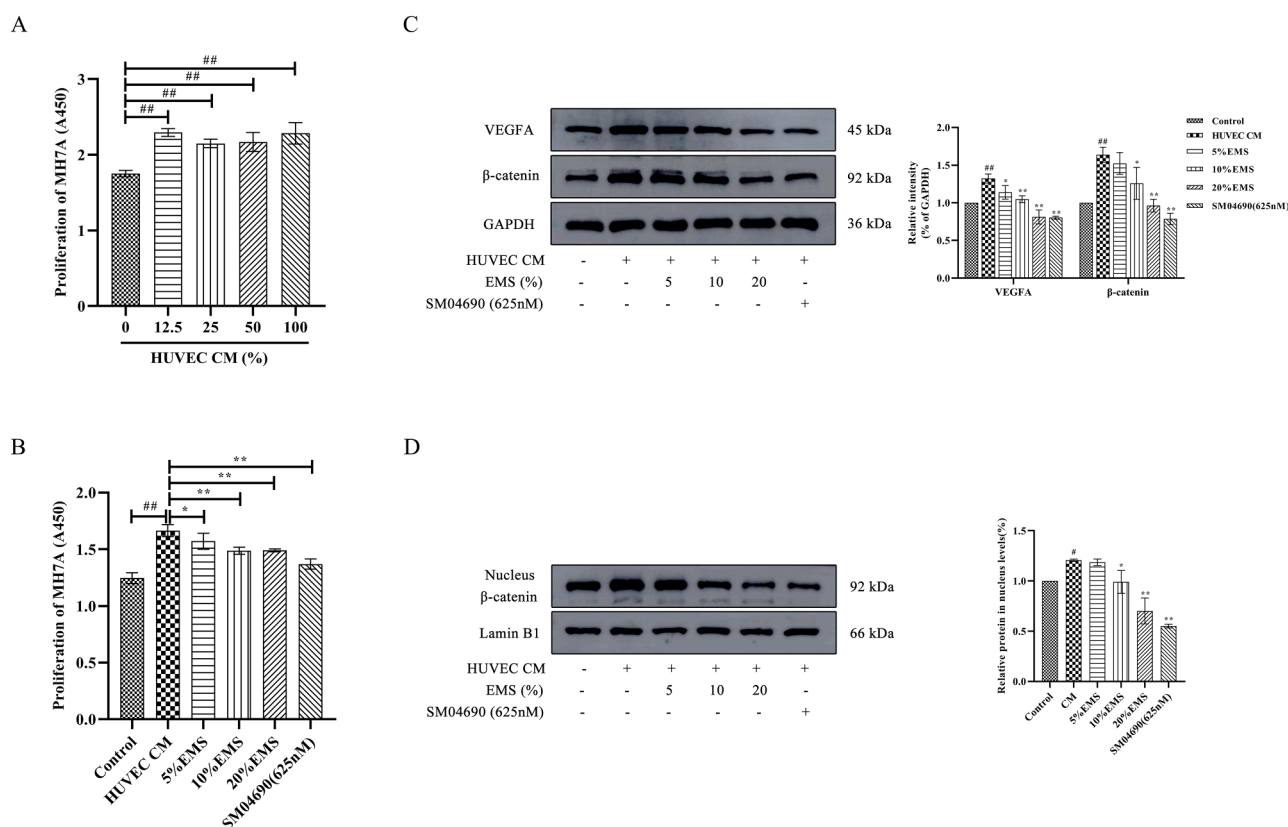


Fig. 4. EMS inhibited crosstalk between MH7A cells and HUVECs and altered the biological functions of MH7A cells. (A) CCK-8 analysis of the proliferation of MH7A cells treated with different concentrations of HUVEC CM for 24 h ($n = 6$). (B) Effects of CM from the different groups on the proliferation of MH7A cells ($n = 6$). The protein levels of (C) β -catenin, VEGFA and (D) β -catenin in the nucleus were determined by western blotting ($n = 3$). The data are presented as the mean $\pm$ SD. $^{\#}p < 0.05$, $^{##}p < 0.01$ vs. the control group; $^{*}p < 0.05$, $^{**}p < 0.01$ vs. the HUVEC CM group.

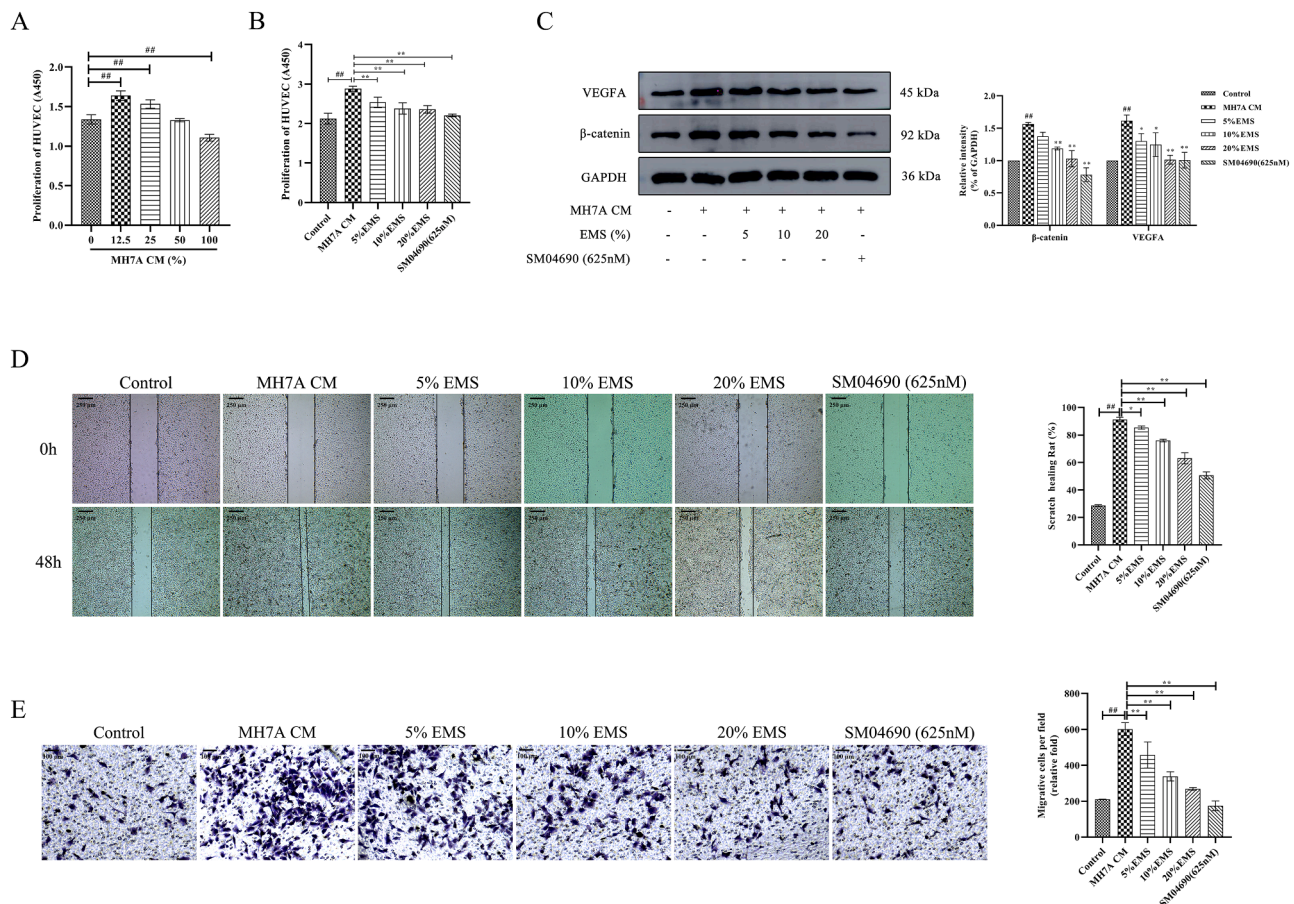


Fig. 5. EMS inhibited crosstalk between MH7A cells and HUVECs and altered the biological functions of HUVECs. (A) CCK-8 analysis of the proliferation of HUVECs treated with different concentrations of MH7A CM for 24 h ($n = 6$). (B) Effects of CM from the different groups on the proliferation of HUVECs ($n = 6$). (C) The protein levels of β -catenin and VEGFA were determined by western blotting ($n = 3$). Cell migration was examined using (D) a scratch assay and a (E) Transwell chamber ($n = 3$). The data are presented as the mean $\pm$ SD. $^{##}p < 0.01$ vs. the control group; $^{*}p < 0.05$, $^{**}p < 0.01$ vs. the MH7A CM group.

significantly reduced paw swelling, the arthritis index and organ indices in AA rats and delayed the progression of AA. Histopathological analysis showed that EMS alleviated the symptoms of synovial proliferation, cellular infiltration and pannus formation in AA rats. Thus, the therapeutic effects of EMS on AA rats were confirmed.

The most typical pathological process of RA involves the accumulation of FLSs in joints, synovial thickening and the secretion of many inflammatory cytokines. Numerous studies have confirmed that abnormal FLS activation plays an important role in the pathogenesis of RA. Many FLSs in joint synovial fluid can excessively migrate, invade and proliferate, ultimately producing proinflammatory cytokines that mediate inflammatory responses and joint destruction (Li et al., 2023). In addition, the synovium is infiltrated by FLSs that abnormally proliferate, undergo insufficient apoptosis and produce cytokines, and angiogenesis continues to occur, resulting in synovial angiogenesis (Sun et al., 2020). IFN- γ is the only member of the type II IFN family that is mainly secreted by immune cells. The production of IFN- γ by activated natural killer T cells (NKT) might be associated with the activation of other immune cells, which could amplify the inflammatory process (Matoba et al., 2009). Yao et al. claimed that IFN- γ -induced production of invariant NKT cells plays an important role in the inflammatory response by activating the ERK signaling pathway (Li et al., 2015). IL-17A is an important inflammatory cytokine in the IL-17 family. Antibody neutralization of IL-17A improves lung function and reduces pulmonary inflammation in animals with silicosis (Cao et al., 2022). Therefore, inhibiting abnormal FLS activation and excessive secretion of proinflammatory cytokines could alleviate the pathological process of RA. Our previous study showed that serum levels of the

anti-inflammatory cytokines IL-10 and TGF- β and the levels of these factors secreted by DCs and peritoneal macrophages were increased in AA rats, while the levels of the proinflammatory cytokines TNF- α , IL-1 β , and IL-23 and the inflammatory mediator PGE2 were decreased after treatment with EMS. In this study, EMS effectively decreased the levels of the proinflammatory cytokines IL-17A and IFN- γ in the peripheral blood of AA rats, thereby reducing the level of inflammation *in vivo* and avoiding further worsening of the disease.

The Wnt/ β -catenin signaling pathway is the crucial signaling axis that regulates cell proliferation, differentiation and migration. Accumulating evidence indicates that the Wnt/ β -catenin signaling pathway is overactivated in the synovium of RA patients, primary FLSs and the synovium of RA experimental animals (Cai et al., 2022; Nakamura et al., 2005; Xiao et al., 2011; S. Zhou et al., 2021). The activation of this signaling pathway is closely related to the pathogenesis of RA. When Wnt/ β -catenin is activated, phosphorylation of GSK-3 β (Ser9) can be induced, resulting in inactivation of GSK-3 β and the ubiquitination of β -catenin. Stable β -catenin accumulates in the cytoplasm and enters the nucleus. After a series of processes, β -catenin can promote the transcription of downstream target genes and participate in a variety of physiological processes. In this study, we found that intra-articular injection of SM04690, an inhibitor of the Wnt/ β -catenin signaling pathway, significantly attenuated synovial hyperplasia and cartilage and bone destruction in AA rats. SM04690 downregulated the secretion of inflammatory cytokines, inhibited the NF- κ B and STAT3 signaling pathways, and reduced the inflammatory response (Deshmukh et al., 2019). The Wnt/ β -catenin signaling pathway, which is an essential regulator of homeostasis and regeneration in tissues, has been proven to

promote abnormal FLS activation. We further found that EMS was an effective antagonist of Wnt/ β -catenin in TNF- α -stimulated FLSs. EMS was shown to reduce Wnt5a/b and APC protein levels, block the phosphorylation of GSK-3 β (Ser9), activate GSK-3 β protein kinase, and reduce the expression of β -catenin and its entry into the nucleus. Similar to the *in vitro* experimental results, EMS significantly inhibited the activation of the Wnt/ β -catenin signaling pathway in the synovium of AA rats. LiCl is a classic GSK-3 β inhibitor that is widely used as an agonist of the Wnt/ β -catenin signaling pathway. As expected, LiCl activated Wnt/ β -catenin in FLSs, and EMS or SM04690 eliminated its effect on FLSs, further suggesting that activating the Wnt/ β -catenin signaling pathway could aggravate the pathogenesis of RA and that EMS could reverse this effect.

VEGFA regulates pathological and physiological angiogenesis, which promotes neovascularization and increases vascular permeability, and its expression is upregulated in many tumours. In this study, we found that EMS decreased the expression of β -catenin in synovial tissue and MH7A cells. Moreover, EMS reversed LiCl-induced activation of β -catenin, resulting in a decrease in VEGFA expression. These results suggested that the downregulation of β -catenin is associated with the inhibition of angiogenesis by EMS and that the antiangiogenic effect of EMS is partially achieved by inhibiting the activation of VEGFA. EMS at a daily dose of 1.5 g/kg inhibited angiogenesis in AA rats. A small animal ultrasound imaging system showed a significant reduction in blood flow in the knee joint in EMS-treated AA rats. The expression of the VEC marker CD31 and smooth muscle cell marker α SMA in synovial tissue was observed by IF staining. The results showed that EMS markedly reduced the number of immature CD31⁺/ α SMA⁺ vessels in AA rats but had no effect on mature CD31⁺/ α SMA⁺ vessels, suggesting that EMS could inhibit the excessive formation of microvessels in the synovial tissue of inflamed joints. bFGF and Ang-2, can promote angiogenesis and are used as prognostic factors. ELISA showed that the levels of bFGF and Ang-2 in the serum of AA rats was significantly decreased after treatment with EMS. These results are consistent with the western blot results

showing that EMS inhibited angiogenesis in AA rats.

The concept of the microenvironment, which was first proposed by Schofield, is used to describe the specific physiological environment for cell survival, which plays an important role in maintaining the stability of the articular cavity in humans and causes a variety of metabolic changes that can initiate an immune response (Bustamante et al., 2017; Li et al., 2019). The inflammatory response is a common immune response following injury and can trigger microenvironmental imbalances. FLSs and VECs are the main members of the inflammatory microenvironment in the RA synovium. Their abnormal function is not conducive to the stability of joint structure and plays an important role in angiogenesis. It has been confirmed that abnormal FLS proliferation results in the release of many inflammatory factors and the infiltration of VECs. VECs are activated in the inflammatory microenvironment, which promotes cell proliferation and migration and induces angiogenesis (Wang et al., 2022). The bidirectional interaction between FLSs and VECs leads to abnormal FLS activation, which is accompanied by the release of inflammatory mediators and signal transduction molecules that are involved in the pathogenesis of RA and joint dysfunction. In this study, we generated a coculture model to simulate the relationship between FLSs and VECs in the synovial microenvironment *in vitro*. The results showed that when MH7A cells and HUVECs were incubated with CM, HUVEC CM significantly promoted the viability of MH7A cells. MH7A CM significantly enhanced the proliferation and migration of HUVECs and the activation of β -catenin and VEGFA, the key targets in the Wnt/ β -catenin signaling pathway. Moreover, EMS reversed these effects, as did SM04690. As summarized in Fig. 6, our results indicated that the crosstalk between MH7A cells and HUVECs could activate the Wnt/ β -catenin signaling pathway, enhance the migration of HUVECs, and participate in the formation of microvessels. EMS inhibited angiogenesis in the inflammatory microenvironment by inhibiting the expression of β -catenin, decreasing the level of VEGFA and weakening the migration of HUVECs. Combined with the results of previous animal experiments, we believe that there are some proangiogenic factors in the

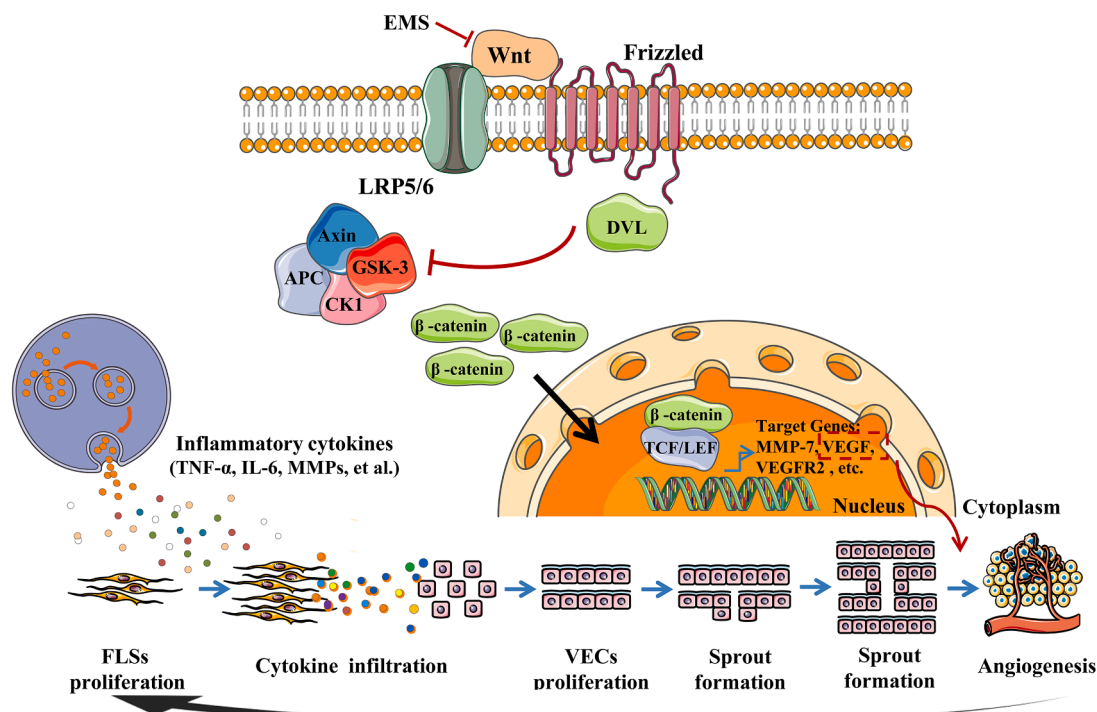


Fig. 6. Schematic illustration of how EMS mediates abnormal FLS activation to inhibit angiogenesis in RA via the Wnt/ β -catenin signaling pathway. The crosstalk between MH7A and HUVEC is involved in the pathogenesis of RA, and the activation of Wnt/ β -catenin signaling pathway mediates angiogenesis. EMS could inhibit angiogenesis in inflammatory microenvironment by suppressing Wnt/ β -catenin signaling pathway, thereby decreasing the expression of β -catenin in the nucleus and downregulating the expression of VEGFA.

inflammatory microenvironment of RA that promote changes in the biological functions of MH7A cells and HUVECs. EMS inhibited angiogenesis in not only AA rats but also the inflammatory RA microenvironment *in vitro*. Taken together, these results suggest that the Wnt/ β -catenin signaling pathway mediates angiogenesis and participates in RA inflammation and that EMS can inhibit Wnt/ β -catenin-mediated angiogenesis, thereby inhibiting RA.

Conclusion

On the one hand, EMS has a therapeutic effect on AA rats, and its mechanism might involve the Wnt/ β -catenin signaling pathway. On the other hand, angiogenesis is involved in the pathogenesis of AA in rats, and EMS can inhibit angiogenesis in AA rats. Furthermore, crosstalk between MH7A cells and HUVECs is involved in the pathogenesis of RA, and Wnt/ β -catenin activation mediates the formation of microvessels. EMS can inhibit angiogenesis in the inflammatory microenvironment by inhibiting the Wnt/ β -catenin signaling pathway, thereby reducing the expression of β -catenin in the nucleus and downregulating the expression of VEGFA. Our results provide a theoretical basis and biological strategy for the use of EMS to improve clinical symptoms in patients with RA.

Author contributions

Simeng Chen: Conceptualization, Methodology, Writing-review & editing. **Jin Wang:** Conceptualization, Writing-review & editing. **Jiayu Wang:** Visualization, Writing-review & editing. **Xiaoyi Jia:** Conceptualization, Methodology, Supervision. **Zihua Xuan:** Conceptualization, Supervision. **Zhiluo Cheng:** Data curation. **Xiangwen Meng:** Methodology. **Wenrui Su:** Data curation. All data were generated in-house, and no paper mill was used. All authors agree to be accountable for all aspects of work ensuring integrity and accuracy.

Supplementary materials

Phytochemical analysis of the EMS extracts by UPLC are shown in Supplementary Fig. 1. (A) EMS extracts. (B) Mixed reference substance. 1: Phellodendrine; 2: Magnoflorine; 3: 7-Hydroxycoumarin; 4: Jatrorrhizine; 5: Berberine; 6: Atractylenolide I.

The total ion chromatograms of EMS extracts in serum samples in positive mode (A) and negative mode (B) are shown in Supplementary Fig. 2.

The identification of the chemical constituents of the EMS extracts in rat serum is shown in Supplementary Table 1.

The arthritis index scoring criteria are shown in Supplementary Table 2.

The inflammation scoring criteria are shown in Supplementary Table 3.

The scoring criteria for synovial proliferation, cellular infiltrate and pannus formation are shown in Supplementary Table 4.

The bone erosion scoring criteria are shown in Supplementary Table 5.

The effect of different concentrations of LiCl on MH7A cells after 24 h is shown in Supplementary Fig. 3. (A) The morphology of MH7A cells treated with different concentrations of LiCl for 24 h was observed by inverted microscopy ($n = 6$). (B) The viability of MH7A cells treated with different concentrations of LiCl for 24 h was examined using the CCK-8 assay ($n = 6$). (C) The protein levels of β -catenin in MH7A cells stimulated with different concentrations of LiCl were determined by western blotting ($n = 3$). The data are presented as the mean $\pm$ SD. * $p < 0.05$, ** $p < 0.01$ vs. 0 LiCl group.

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.phymed.2023.155064](https://doi.org/10.1016/j.phymed.2023.155064).

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